

Design FMEA - VBF-T5R1-DFMEA-01

VBF-T5R1 delivery package - generated 2026-08-25

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Design FMEA (qualitative version, based on simulation signals) - VBF-T5R1-DFMEA-01

Case: t5_r1, design R7B. Caliber annotation: qualitative version - severity/occurrence ratings are three-level (high/medium/low) based on the magnitude of simulation values deviating from thresholds; RPN uses a simplified S?O matrix. Complete FMEA (process/supplier failures) is N/A beyond the pure-simulation boundary.

#	Failure mode	Failure cause	Simulation signal (detectability basis)	Severity	Occurrence	Mitigation in design
1	Negative electrode lithium plating at 4C fast charge	Anode surface potential driven below 0 V by end-of-charge solid-phase diffusion overpotential + electrolyte transport limits	`anode_potential_v` min: R7B = +0.0183 V (Chen) / +0.0180 V (OKane) vs threshold 0 V - margin only +18 mV	high	low-medium (margin is thin; eroded by thicker/deeper charge or coarser graphite)	2.61 ?m negative particles (? = R2/Ds ~ 210 s ? 800 s charge); ? 2.0 S/m, D_e 4.5e-10, t? 0.45, ?_neg 0.45, sep 8 ?m; N/P 1.40
2	Thermal runaway risk (temperature exceeds 60 ?C red line at 4C/45 ?C)	Heat generation exceeds cooling capacity	`T_max_K` = 326.8 K (Chen) / 328.2 K (OKane) vs 333.15 K - margin ~5?6 K	high	low at h = 60 W/(m2*K) (immersion-class); rises sharply if cooling degrades (h=40 failed at 338.8/341.0 K in R4)	h = 60 W/(m2*K) is a hard thermal-management requirement; cooling-area 0.00531 m2
3	Electrolyte oxidative decomposition (voltage window)	Electrolyte HOMO above cathode potential at high SOC	Not simulated in this case (true DFT endorsement skipped, real_compute=false; no IE/EA numbers claimed)	medium	low-medium (standard EC-based + LiPF6 at 4.2 V; no signal to the contrary, but unverified)	Keep 4.2 V upper cutoff; formulation validation before production
4	Insufficient capacity	Cathode/anode usable capacity below requirement	1C `capacity_ah` = 6.210/6.209 Ah vs 5 Ah nominal - exceeds nominal (anode-unlocked, N/P 1.40)	low-medium	low	Anode-limited 1C discharge confirmed (R4C evidence); N/P 1.40 margin
5	Internal short circuit via separator	8 ?m separator (thinned for ionic transport) + particle penetration	N/A - no simulation protocol for short circuits	high	low-medium (thin-separator class; production-validated 6?9 ?m separators exist, but QC critical)	Separator integrity/particle-size QC; 8 ?m chosen over 6 ?m (R7D showed no benefit)
6	Plating-margin erosion in manufacturing	Tolerance drift in particle size/porosity/thickness away from R7B values	Sensitivity known from sweep: neg radius 3.5 ?m -> +6 mV margin; 5.86 ?m -> ?19 mV (plating, R6D)	high	low-medium	Tolerances on negative particle radius (2.61 ?m class), ?_neg 0.45, sep 8 ?m specified as design-critical

Conclusion

Highest-risk items: #1 plating (margin +18 mV, high severity) and #2 thermal (margin ~5?6 K at h=60) - both are mitigated in the design and both PASS under dual-system DFN simulation, but their margins are thin by engineering standards and are called out as the primary experimental-validation targets. Items #3/#5 rely on no simulation signal and are flagged for physical testing. Complete FMEA (process/supplier failures): N/A (beyond pure simulation boundary).